



西北工业大学

Lab report

Structural Characterisation

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Experiment	Phase analysis by XRD

Experiment I Phase analysis by XRD

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Grade	Third				
Contents	1. Working principle of X-ray diffractometers 2. Calibration of XRD spectrum by software 3. Calibration of XRD spectrum by yourself				

1. Working principle of X-ray diffractometers (30 marks)

The X-ray diffractometer (XRD) is a key instrument for phase identification and structural characterization of crystalline materials. Its operation is governed by Bragg's Law, $n\lambda = 2d\sin \theta$, where λ is the wavelength of the incident X-rays, d is the interplanar spacing, and θ is the incident angle relative to the crystal planes. Only when this condition is satisfied does constructive interference occur, producing a detectable diffracted beam.

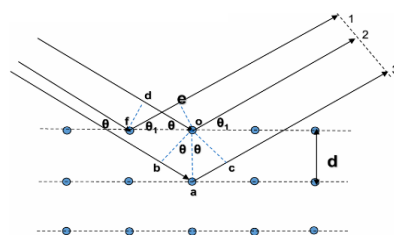


Figure 1. Bragg's Law illustration

The XRD system is composed of three essential components: the X-ray generator, the goniometer, and the detector. The X-ray tube produces monochromatic radiation that is collimated and directed onto the specimen. The goniometer simultaneously rotates the sample and detector at a fixed 1:2 angular ratio, maintaining a constant geometric relationship so that the detector records diffracted beams at 2θ . This configuration ensures that only X-rays diffracted from crystal planes parallel to the sample surface contribute to the measured signal.

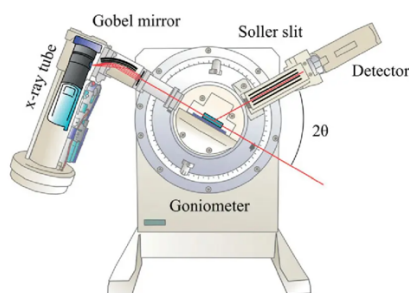


Figure 2. Work diagram of X-ray diffractometer

During measurement, the detector scans along the focusing circle and records diffracted intensity as a function of 2θ , generating the characteristic diffraction pattern. From this pattern, information such as lattice parameters, phase composition, and crystallographic texture can be extracted. Common scan modes include continuous scanning for rapid phase identification and step scanning for high-precision intensity measurements required in quantitative analysis.

In essence, XRD provides a non-destructive method for probing periodic atomic structures through constructive interference of X-rays, making it an indispensable tool in materials science, chemistry, and geology.

2. Calibration of XRD spectrum by software. (30 marks)

Single phase: 7.mdi

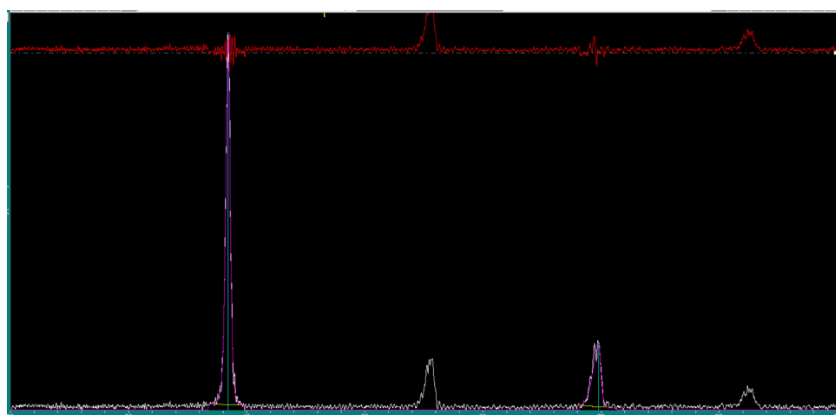


Figure 3. Collecting diffraction data of 7.mdi

#	2-Theta	d(?)	BG	Height	I%	Area	I%	FWHM
1	38.386	2.3431	8	383	100.0	9671	100.0	0.429
2	38.948	2.3105	4	35	9.1	379	3.9	0.184
3	55.249	1.6612	3	43	11.2	2142	22.1	0.847
4	55.466	1.6553	4	48	12.5	2020	20.9	0.715
5	55.709	1.6486	6	47	12.3	1913	19.8	0.692
6	69.111	1.3580	3	32	8.4	1401	14.5	0.744
7	69.462	1.3520	5	64	16.7	2774	28.7	0.737

Figure 4. Peak-seeking results of 7.mdi

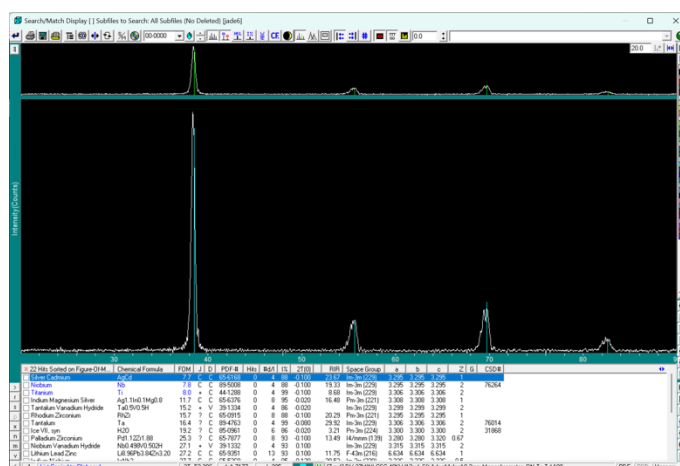


Figure 5. Element Retrieval of 7.mdi

Double phase: 11.mdi

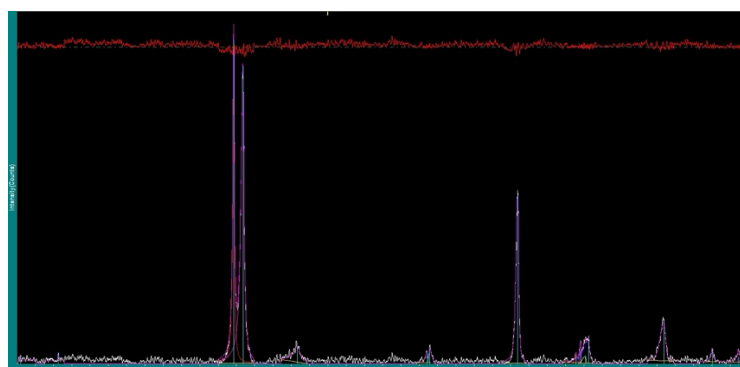


Figure 6. Collecting diffraction data of 11.mdi

Peak Search Report (30 Peaks, Max...)									
Close	Print	Save	Copy	Erase	Customize	Rescale	Help		
#	2-Theta	d(?)	BG	Height	I%	Area	I%	FWHM	
1	20.329	4.3647	3	54	2.4	371	0.9	0.134	
2	24.525	3.6267	17	54	2.4	344	0.8	0.124	
3	25.195	3.5318	14	32	1.4	197	0.5	0.120	
4	26.036	3.4195	23	40	1.8	877	2.1	0.427	
5	27.574	3.2322	11	40	1.8	478	1.1	0.233	
6	28.675	3.1105	12	26	1.2	265	0.6	0.198	
7	29.504	3.0250	16	34	1.5	215	0.5	0.123	
8	34.663	2.5857	17	32	1.4	107	0.3	0.065	
9	35.475	2.5284	12	33	1.5	488	1.1	0.288	
10	37.054	2.4242	9	29	1.3	157	0.4	0.105	
11	41.110	2.1939	12	29	1.3	176	0.4	0.118	
12	43.688	2.0702	25	2224	100.0	31761	74.3	0.278	
13	44.677	2.0266	15	1952	87.8	42755	100.0	0.427	
14	49.752	1.8311	42	55	2.5	579	1.4	0.205	
15	50.281	1.8131	42	55	2.5	980	2.3	0.347	
16	50.607	1.8022	39	104	4.7	2423	5.7	0.454	
17	54.030	1.6958	10	35	1.6	138	0.3	0.077	
18	54.431	1.6843	10	35	1.6	281	0.7	0.156	
19	54.908	1.6708	9	25	1.1	345	0.8	0.269	
20	55.507	1.6541	10	38	1.7	575	1.3	0.295	
21	56.345	1.6315	14	33	1.5	384	0.9	0.227	
22	56.567	1.6256	13	29	1.3	253	0.6	0.170	
23	60.749	1.5233	6	40	1.8	584	1.4	0.284	
24	62.719	1.4802	11	20	0.9	166	0.4	0.162	
25	64.240	1.4487	17	27	1.2	102	0.2	0.074	
26	64.811	1.4373	8	77	3.5	1263	3.0	0.319	
27	65.191	1.4299	14	111	5.0	1674	3.9	0.294	
28	68.021	1.3771	9	23	1.0	349	0.8	0.296	
29	68.719	1.3648	10	19	0.9	231	0.5	0.237	
30	69.270	1.3553	9	21	0.9	277	0.6	0.257	

Figure 7. Peak-seeking results of 11.mdi

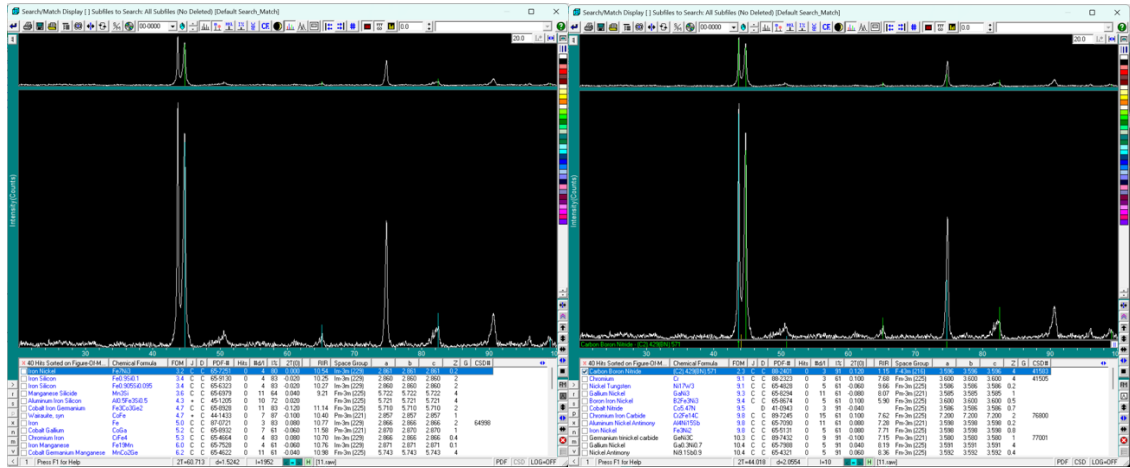


Figure 8. Element Retrieval of 11.mdi

3. Calibration of XRD spectrum by yourself. (40 marks)

According to Bragg's equation and equation of distance between planes in cubic lattice:

$$2d \sin \theta = \lambda$$

$$\sin^2 \theta = \frac{\lambda^2}{4a^2} (H^2 + K^2 + L^2)$$

Without considering the constant part, the series could be written as:

$$\sin^2 \theta_1 : \sin^2 \theta_2 : \dots = (H_1^2 + K_1^2 + L_1^2) : (H_2^2 + K_2^2 + L_2^2) : \dots$$

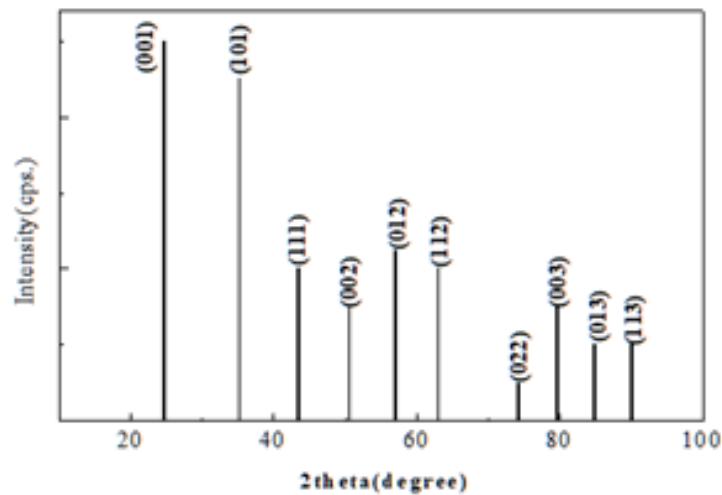


Figure 9. Intensity and position diagram of sample 1

Table 1. Calibration of XRD spectrum

Diffraction peak	Intensity	2θ	θ	sin ² θ	sin2θ _i /sin2θ ₁	Possible diffraction planes	N ₁ /N ₂
1	highest	24	12	0.0432	1	(0 0 1)	1
2	high	35	17.5	0.0904	2.0926	(1 0 1)	2
3	medium	43	21.5	0.1343	3.1088	(1 1 1)	3
4	medium	50	25	0.1786	4.1342	(0 0 2)	4
5	medium	58	29	0.235	5.4398	(0 1 2)	5
6	medium	62	31	0.2653	6.1412	(1 1 2)	6
7	weakest	74	37	0.3622	8.3842	(0 2 2)	8
8	medium	80	40	0.4132	9.5648	(0 0 3)	9
9	weak	85	42.5	0.4564	10.5648	(0 1 3)	10
10	weak	90	45	0.5	11.574	(1 1 3)	11

In this case, sin²θ₁: sin²θ₂: sin²θ₃: sin²θ₄: sin²θ₅: sin²θ₆: sin²θ₇: sin²θ₈: sin²θ₉: sin²θ₁₀=1:2:3:4:5:6:8:9:10:11, which conforms to the ratio of sin²θ in cubic crystal system, so that the crystal structure of the sample is **cubic crystal system**.

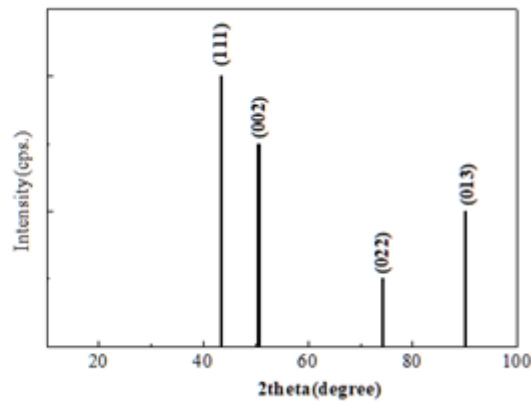


Figure 10. Intensity and position diagram of sample 2

Table 2. Calibration of XRD spectrum

Diffraction peak	Intensity	2θ	θ	sin ² θ	sin2θ _i /sin2θ ₁	Possible diffraction planes	N ₁ /N ₂
1	highest	42	21	0.1284	1	(1 1 1)	3
2	high	50	25	0.1786	1.3909	(0 0 2)	4
3	weakest	74	37	0.3622	2.8209	(0 2 2)	8
4	medium	90	45	0.5000	3.8941	(1 1 3)	11

In this case, sin²θ₁: sin²θ₂: sin²θ₃: sin²θ₄=3:4:8:11, which conforms to the ratio of sin²θ in face-centered cubic, so that the crystal structure of the sample is **face-centered cubic**.

